

SOP #18 – ATI (Analysis to Improve) for Plumbing Systems

Goal: Increase reliability and reduce water waste through predictive maintenance and material optimization

1. Performance Metrics

Metric	Benchmark	Strategy for Improvement
Leak Incidence Rate	<3% of monthly checks	Install leak sensors, automate logs
Hot Water Delivery Time	<15 seconds to fixture	Add recirculation pump with timer
Annual Backflow Test Compliance	100% complete & certified	Maintain calendar & certified vendor list
Water Pressure Variation	<10 psi during peak use	Rebalance PRVs and storage tanks

2. Optimization Strategies

- Install low-flow fixtures and valves to reduce water use
- Use PEX piping with expansion fittings to reduce maintenance
- Apply insulating sleeves on all domestic hot water lines
- Automate water heater diagnostics with smart modules
- Implement greywater recovery systems for irrigation or flushing

3. Legal & Safety Compliance

- UPC/IPC Plumbing Code – Fixture counts, slopes, pipe sizing
- OSHA 1910.141 – Sanitation and potable water access
- EPA Safe Water Act – Backflow prevention, water quality tests
- ASSE Standards 1013/1015 – Backflow device compliance
- State Licensing & Inspections – Document all repairs & permits